AQUATIC MAMMALS

Three groups of mammals have adapted to aquatic life, developing a streamlined body and the ability to stay underwater for long periods (although all of them return to the surface to breathe). The largest group, the whales and dolphins, are the most specialized. They have lost their body hair and spend their entire life in water. Like whales, seals and sea lions rely on subcutaneous fat to keep warm, but have retained their fur, which is kept waterproof by an oily secretion from the sebaceous glands. Sirenians (the manatees and dugong) live in warm coastal waters and estuaries and are the only herbivorous aquatic mammals. The sea otter is the only other mammal that spends most of its life in water. It lacks subcutaneous fat and instead keeps itself warm by trapping air in its dense fur.

LIVING IN THE SEA
Aquatic mammals, such as this humpback whale, evolved from land-living mammals. Although the back limbs have been lost, some whales retain a vestigial pelvic girdle.

GRACEFUL MOVERS
Seals, such as this harbor seal, are highly acrobatic swimmers, with front and back limbs that have been modified into webbed flippers. Ungainly on land, they rarely venture far from the water, even to give birth to their young.

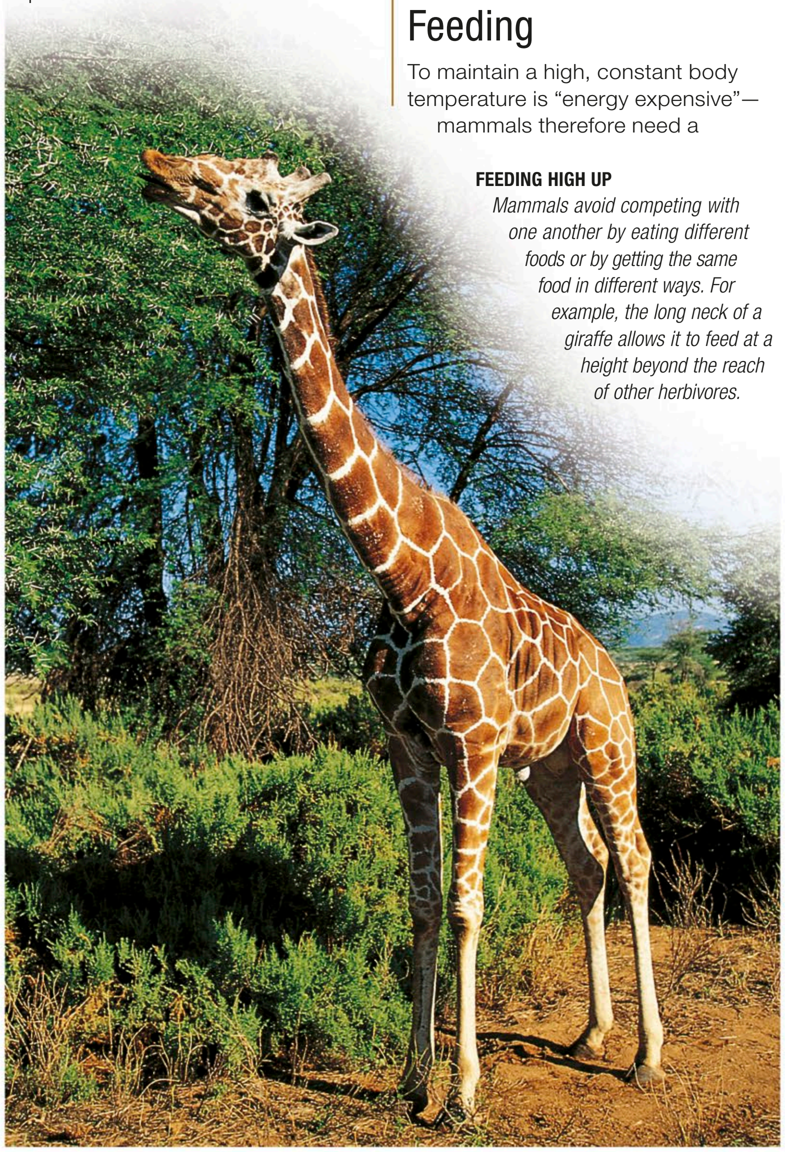
spreading their arms to expose their thinly haired undersurface.

Behavioral patterns also help regulate body temperature. For example, in the desert and in tropical grasslands, in the heat of the day rodents retreat into cool, moist burrows, while larger mammals rest in the shade in cool depressions in the ground. Body coloring is another important factor: dark colors absorb heat; light colors reflect it. Desert mammals are therefore often light fawn in color, while those living in cool climates are dark. This arrangement may conflict with the need for camouflage—where there is winter snow, mammals may turn white (stoats) or have a permanently white coloration (polar bears), and possess extra-thick fur to compensate.

Feeding

To maintain a high, constant body temperature is "energy expensive"—mammals therefore need a nutritious and plentiful diet. While the earliest mammals were probably predators, different species have since adapted to meet their dietary requirements in a variety of ways. Some eat animal prey—this is a carnivorous diet (and includes insect-eating). Other mammals, called herbivores, eat plants. A herbivorous diet includes sub-types such as fruit-eating and grass-eating. An omnivore eats both animal prey and plants.

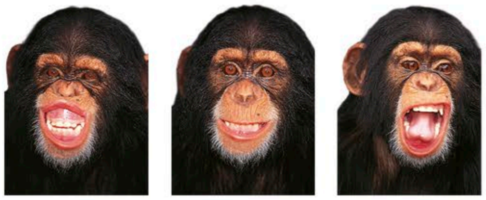
Carnivorous mammals have a short, simple digestive tract, because the proteins, lipids, and minerals found in meat require little in the way of specialized digestion. Plants, on the other hand, contain complex carbohydrates, such as cellulose. The digestive tract of a herbivore is long and bulky and hosts bacteria that ferment these substances and make them available for digestion. The bacteria are housed either in a multichambered stomach or in a large cecum (a blind-ending sac in the large intestine). The size of an animal is a factor in determining diet type. Small mammals have a high ratio of heat-losing surface area to heat-generating volume, which means that they tend to have high energy requirements and a high metabolic rate. Because they cannot tolerate the slow, complex digestive process of a herbivore, mammals that weigh less than about 18 oz (500 g) are mostly insectivorous. Larger mammals, on the other hand, generate more heat and less of this heat is lost. They can therefore tolerate either a slower collection process (those that prey on large vertebrates) or a slower digestive process (herbivores). Furthermore, mammals that weigh more than about 18 oz (500 g) usually cannot collect enough insects during their waking hours to sustain themselves. The only large insectivorous mammals are those that feed on huge quantities of colonial insects (ants or termites).



FEEDING HIGH UP
Mammals avoid competing with one another by eating different foods or by getting the same food in different ways. For example, the long neck of a giraffe allows it to feed at a height beyond the reach of other herbivores.

Social structures

Mammals communicate socially by scent, either from glands (which may be located in the face, in the feet, or



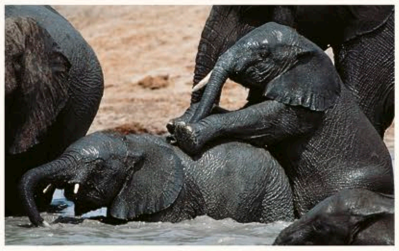
FEAR | SUBMISSION | EXCITEMENT

FACIAL EXPRESSION
Some mammals communicate using facial expressions. This ability is well developed in primates, such as the chimpanzee. Fear is shown by baring the teeth, submission by a pouting smile, and excitement by exposed teeth and an open mouth.

PLAY

A great deal of a young mammal's learning occurs through play, when infants experiment with adult behaviors, such as fighting and hunting. Play can also take the form of exploring and displaying to one another. Hoofed mammals, such as deer, establish dominance rankings when young to avoid conflict as adults. This reduces the risk of serious injury that would make them vulnerable to predators. Predators must learn to stalk and kill prey to survive as adults.

SURVIVAL SKILLS
Potential predators, such as these lion cubs, use play to practice pouncing, biting, and raking with the back feet, stopping before real damage is done.



FINDING THEIR PLACE
Like many herbivores, these elephant calves use play to establish their rank in the herd. Young elephants also use their trunks to examine novel objects.